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Thiol-bearing synthetic peptides retain the antioxidant activity of apolipoproteinA-I_{Milano}

Zhen Jia, Pradeep Natarajan, Trudy M. Forte, and John K. Bielicki*

Department of Genome Sciences, Division of Life Sciences, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA

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Abstract

Apolipoprotein(apo)A-I_{Milano} (R173C) and apoA-I_{Paris} (R151C) are rare cysteine variants of wild-type (WT) apoA-I that possess novel antioxidant properties on phospholipid surfaces. Yet, the two variants differ in their ability to inhibit lipid peroxidation. In this study, we used synthetic peptides (18 mers) to investigate the structural basis for the difference in antioxidant activity between apoA-I_{Milano} and apoA-I_{Paris}. A peptide (aa 167-R173C-184) based on the amphipathic α helix harboring the R173C mutation inhibited superoxide anion-mediated oxidation of phospholipid in a dose-dependent manner, but it failed to directly quench superoxide anions in aqueous solution, indicating that the peptide acted at the level of phospholipid to inhibit lipid peroxidation just like the full-length cysteine variant. Peptide 145-R151C-162 based on the helical segment containing R151C exhibited the same capacity as peptide 167-R173C-184 to inhibit lipid peroxidation. Thus, the difference in antioxidant activity between apoA-I_{Milano} and apoA-I_{Paris} was not governed by the primary amino acid sequence of their individual amphipathic α helices, rather contextual constraints within the full-length variants set the difference in antioxidant activity. Cysteine-free peptides were weak inhibitors of lipid peroxidation. These results suggest that thiol-bearing helical peptides based on apoA-I_{Milano} may be useful to combat inflammatory related diseases. © 2002 Elsevier Science (USA). All rights reserved.

Keywords: ApolipoproteinA-I; ApolipoproteinA-I_{Milano}; ApolipoproteinA-I_{Paris}; Amphipathic α helix; Synthetic peptides; Antioxidant activity; Reactive oxygen species; Superoxide anions; Lipid peroxidation; Oxidized phospholipid

Elevated plasma concentrations of HDL-cholesterol are associated with reduced risk of cardiovascular disease (CVD) [1–3]. Yet, a number of rare mutations in apolipoprotein(apo)A-I, the major protein of HDL, produce HDL deficiency in the absence of CVD in humans [4]. ApoA-I_{Milano} (R173C) and apoA-I_{Paris} (R151C) are well-known cysteine variants of apoA-I that produce mild hypertriglyceridemia in addition to HDL deficiency [5–8]. The absence of CVD in human carriers of apoA-I_{Milano} and apoA-I_{Paris} suggests that the cysteine substitutions may confer beneficial properties that play a role in protecting against atherosclerosis.

The presence of a cysteine residue permits disulfide bond formation, allowing apoA-I_{Milano} to form homodimers as well as heterodimers with apoA-II [6]. The same is true for apoA-I_{Paris} [9]. Dimerization of the cysteine variants restricts HDL particle size accounting for the predominance of small, dense HDL_3 sized particles in human plasma [10,11]. The dimeric form of apoA-I_{Milano} on the surface of small, reconstituted HDL particles is thought to enhance cholesterol efflux from cells via a facilitated aqueous diffusion mechanism [12]. Yet, monomers and heterodimers of apoA-I_{Milano} act normally, compared to apoA-I_{WT}, promoting cholesterol efflux from cells via the specific apolipoprotein-mediated efflux pathway [13,14].

The monomeric forms of apoA-I_{Milano} and apoA-I_{Paris} account for a small portion (~10–30%) of the total plasma pool of apoA-I [10,15]. Monomers of apoA-I_{Milano} are cleared more rapidly from the circulation compared to their dimeric counterparts, which may explain why the dimeric form predominates in human carriers [16]. The monomeric form of apoA-I_{Milano} possesses a cysteine residue that is not occupied in the form of a disulfide bridge; thus, the thiol is ''free'' to participate in reaction that may protect against atherosclerosis. Thiol groups in proteins are relatively strong nucleo-

* Corresponding author. Fax: 1-510-486-4750.
E-mail address: jkbielicki@lbl.gov (J.K. Bielicki).

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philes often conferring antioxidant activity. It was recently discovered that the monomeric forms of apoA-I_{Milano} and apoA-I_{Paris} possessed an antioxidant activity distinct from that of apoA-I_{WT}, although the two variants differed in their ability to inhibit oxidative events [14].

ApoA-I possesses a series of 11- and 22-amino acid repeats that form amphipathic α helices [17]. The helical repeats bearing R173C and R151C are located in the C-terminal lipid-binding domain of the protein defined by exon 4 of the apoA-I gene. The R173C mutation is located in helix 7, corresponding to amino acids (aa) 165–186 in apoA-I primary sequence. R151C is located in helix 6, corresponding to aa 143–164. Both stretches of amino acids form class A amphipathic α helices that bind to phospholipid surfaces [17]. The free thiol in helix 7 of apoA-I_{Milano} and helix 6 of apoA-I_{Paris} is similarly positioned at the polar/nonpolar interface of their respective amphipathic α helices [14,18]. Thus, it is not clear as to why the two cysteine variants differ in their ability to inhibit lipid peroxidation.

In this study, we asked whether the individual amphipathic α helical segments harboring the R173C and R151C mutations retained the antioxidant activity of the full-length cysteine variants. This represents an important first step in the development of potential therapeutic peptides that exhibit the newly discovered antioxidant activity of apoA-I_{Milano}. We also used synthetic peptides to gain insight into the underlying basis for the difference in antioxidant activity between apoA-I_{Milano} and apoA-I_{Paris}.

Materials and methods

Materials. 1-Palmitoyl-2-linoleoylphosphatidylcholine was purchased from Avanti Polar Lipids (Alabaster, AL). Xanthine oxidase was from Roche Diagnostics (Mannheim, Germany). The following reagents were obtained from Sigma (St. Louis, MO): soybean lipoxygenase, superoxide dismutase, and cytochrome *c*. Radioisotopes were purchased from Amersham Pharmacia (Buckinghamshire, England) as were the His-trap chelating columns used in the isolation of recombinant apolipoproteins.

Recombinant apolipoproteins. ApolipoproteinA-I_{Milano} and apoA-I_{Paris} were prepared by primer directed mutagenesis as previously described [14,19]. Mutagenized cDNAs were subcloned into a pET-20b bacterial expression vector and plasmids transformed into *Escherichia coli* strain BL21 (DE3). Expressed proteins were purified by His-trap chelating columns (Pharmacia). Recombinant apoA-I_{WT} as well as the two cysteine variants were greater than 95% pure as judged by SDS–PAGE and Coomassie R-250 staining (data not shown). Moreover, SDS–PAGE revealed that apoA-I_{Milano} and apoA-I_{Paris} were isolated in principally (approx. 90%) monomeric form as previously reported [14,19].

Peptide mimetics. Synthetic peptides (Biosynthesis, Lewisville, TX) based on amphipathic α helices 6 (aa 145–162) and 7 (aa 167–184) of apoA-I were composed of the following sequences of amino acids: GEEMRDRARAHVDALRTH, and SDELRQRLAARLEALKEN, respectively. The underlined arginine residues were replaced with cysteine residues to mimic the helices found in apoA-I_{Paris} and apoA-I_{Milano}, respectively. All peptides were modified by N-terminal acetyl group and C-terminal amide group and were dissolved in sterile, deoxygenated Tris (20 mM) buffered (pH 7.4) saline-EDTA (2.7 mM) and stored at 4 °C. Thiol-bearing peptides stored in this manner retained antioxidant activity over a 3-month period.

Assessment of antioxidant activity. A micelle substrate composed of 1-palmitoyl-2-linoleoylphosphatidylcholine dispersed in borate (pH 9.0)/saline-EDTA and deoxycholate was exposed to either soybean lipoxygenase (5 U/μl) or xanthine (0.2 mM)/xanthine oxidase (20 U/ml) to initiate lipid peroxidation as previously described [14,20]. Lipid peroxidation was monitored by ultraviolate absorption spectroscopy at 234 nm (i.e., conjugated diene formation). The mass of phospholipid hydroperoxides was calculated using the molar absorptivity coefficient ($\varepsilon = 29{,}500\,L/(cm\,mol)$) of conjugated dienes. In some experiments, cytochrome *c* was used instead of the phospholipid micelles to determine if the peptide mimetics directly quenched reactive oxygen species (ROS) generated via xanthine/xanthine oxidase in aqueous solution [14].

Cellular cholesterol efflux. J774 macrophages were used to assess the capacity of the synthetic peptides to promote cholesterol efflux. Cells were plated at an initial density of 1×10^6 cells/ml/well in 24-well culture plates and allowed to grow (48 h) to confluent monolayers in RPMI medium containing 1% fetal bovine serum (FBS) and 1 μCi/ml/well [^{14}C]cholesterol [14]. Synthetic peptides in serum-free RPMI were filter sterilized (0.45 μm) and added to cells for the times shown in the figures. Aliqouts of efflux medium were removed and cellular debris was pelleted by centrifugation (1000*g*, 8 min at 4 °C). Radioactivity in medium supernatant was quantified and efflux was expressed as a percentage of the cellular [^{14}C]cholesterol released into the medium as a function of time.

LCAT activity. The capacity of the synthetic peptides to support cholesterol esterification was examined using a recombinant LCAT protein purified from conditioned medium obtained from CHO-K1 cells stably transfected with LCAT cDNA. The recombinant LCAT protein was >95% pure and exhibited a molecular weight of 66 kDa as judged by SDS–PAGE (data not shown). Cholesterol esterification was assessed using an exogenous substrate procedure [21]. Proteoliposomes composed of the cysteine containing peptides were prepared using egg-yolk phosphatidylcholine (PC) and unesterified cholesterol (16:250:12.5, mole ratios, respectively). Proteoliposomes were exposed to 20 μg/ml purified LCAT protein in reaction mixtures consisting of human serum albumin (0.5%), β-mercaptoethanol (5 mM), Tris (20 mM) buffered saline EDTA (2.7 mM). Incubations were performed at 37 °C for 1 h. Results are expressed as a percentage of the initial [^{14}C]cholesterol converted to [^{14}C]cholesteryl ester.

Results

Fig. 1A shows that apoA-I_{Milano} was 2-fold more potent than apoA-I_{Paris} in protecting phospholipid from oxidation induced water-soluble reactive oxygen species (ROS), i.e., superoxide anion and hydroxyl radicals. At 50 μg protein/ml, apoA-I_{Milano} inhibited lipid peroxidation by 50% while a similar dose of apoA-I_{Paris} provided 25% protection. Previous studies using lipoxygenase to initiate lipid peroxidation revealed that apoA-I_{Milano} was consistently more effective than apoA-I_{Paris} over a wide range of concentrations [14]. Wild-type (WT) apoA-I failed to inhibit lipid peroxidation induced by ROS, indicating that the antioxidant activity of apoA-I_{Milano} and apoA-I_{Paris} was related to the presence of the free thiol in the monomeric form of the variants. Fig. 1B

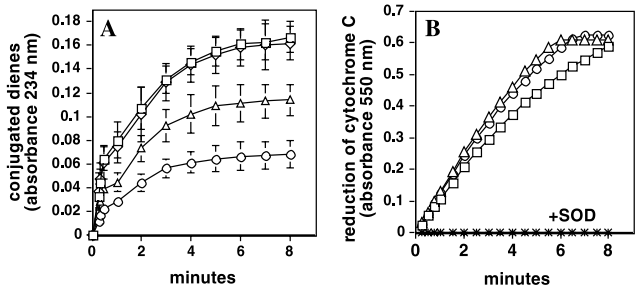


Fig. 1. Capacity of apoA-I_{Milano} and apoA-I_{Paris} to inhibit lipid peroxidation. (A) Phospholipid micelles were exposed to superoxide anions generated by xanthine (0.2 mM)/xanthine oxidase (20 U/ml) in the absence (squares) and presence of wild-type apoA-I (diamonds), apoA-I_{Paris} (triangles), and apoA-I_{Milano} (circles). The concentration of each of the apolipoproteins was 50 μg/ml. Conjugated dienes are means ± SD, $n = 3$. (B) Reduction of cytochrome *c* mediated by superoxide anions. Symbols are the same as in (A). SOD = superoxide dismutase (300 U/ml).

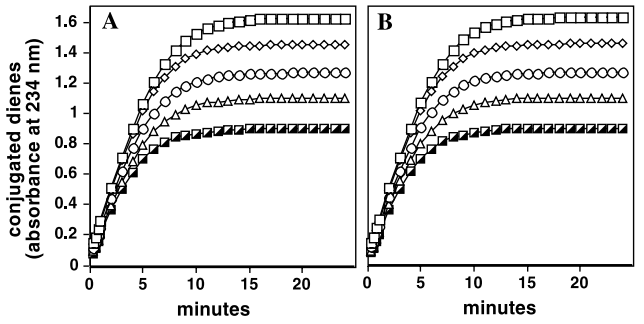


Fig. 3. Capacity of peptides 167-R173C-184 and 145-R151C-162 to inhibit lipoxygenase-mediated lipid peroxidation. Phospholipid micelles were exposed to soybean lipoxygenase (5 U/μl) in the absence (squares) and presence of increasing concentrations of peptides 167-R173C-184 (A) and 145-R151C-162 (B). Diamonds, circles, triangles, and half-darkened squares correspond to 100, 200, 300, and 400 μg/ml of each peptide. Results are representative of three identical experiments.

shows that both cysteine variants were unable to prevent the reduction of cytochrome *c* mediated by ROS indicating that apoA-I_{Milano} and apoA-I_{Paris} could not directly quench superoxide anions in the aqueous phase. This suggests that apoA-I_{Milano} and apoA-I_{Paris} acted at the level of phospholipid to inhibit lipid peroxidation mediated by superoxide anions.

In contrast to the difference in antioxidant activity between the two full-length cysteine variants, synthetic peptides derived from the individual amphipathic α helices possessing the R173C and R151C mutations acted identically, exhibiting the same capacity to inhibit lipid peroxidation. The results in Fig. 2A show that peptide 167-R173C-184 based on helix 7 of apoA-I_{Milano} inhibited lipid peroxidation in a dose-dependent manner, reaching 50% protection at 200 μg/ml. A similar level of protection was obtained using peptide 145-R151C-162 based on helix 6 of apoA-I_{Paris} (Fig. 2B), indicating that the difference in antioxidant activity between the two full-length cysteine variants as shown in Fig. 1 was not due to differences in the primary amino acid sequence of the individual helical segments harboring the cysteine mutations. Both thiol-bearing peptides were unable to directly quench water-soluble ROS in lipid-free form as judged using the cytochrome *c* protection assay as shown in Fig. 2C. This indicates that the antioxidant activity of the synthetic peptides was directed toward phospholipid just like the full-length cysteine variants. Figs. 3A and B show that peptides 167-R173C-184 and 145-R151C-162 inhibited lipoxygenase-mediated oxidation of phospholipid in dose-dependent manner. The peptides behaved similarly providing the same level of protection at each dose.

As can be seen in Fig. 4, peptide 145-R151C-162 based on apoA-I_{Paris} inhibited lipid peroxidation when added before and after initiation of oxidation as previously observed using the monomeric form of apoA-I_{Milano} [14]. Similar results were obtained using

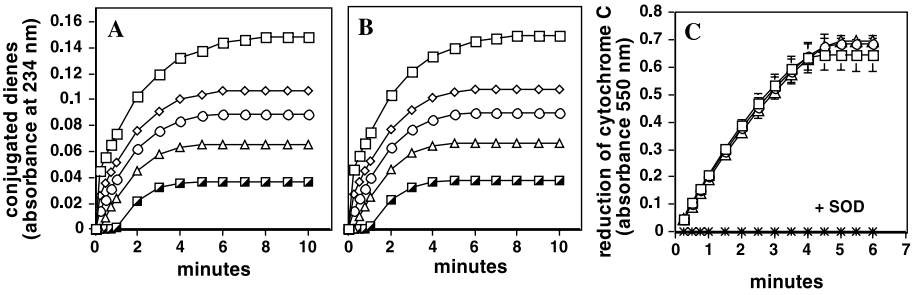


Fig. 2. Antioxidant activity of peptide mimetics of apoA-I_{Milano} and apoA-I_{Paris}. (A, B) Phospholipid micelles were exposed to superoxide anions generated by xanthine (0.2 mM)/xanthine oxidase (20 U/ml) in the absence (squares) or presence of peptides 167-R173C-184 (A) and 145-R151C- 162 (B). Diamonds, circles, triangles, and half-darkened squares correspond to 100, 200, 300, and 400 μg/ml of each peptide. Results are representative of three identical experiments; the maximum absorbance values obtained at 10 min averaged 0.173 ± 0.025 for controls (i.e., no peptide), and 0.097 ± 0.014 and 0.097 ± 0.027 for peptide 167-R173C-184 and 145-R151C-162 each at 200 μg/ml, respectively (means ± SD, $n = 3$). (C) Reduction of cytochrome *c* mediated by superoxide anions; squares, no peptides; circles, peptide 167-R173C-184 (200 μg/ml); triangles, peptide 145-R151C-162 (200 μg/ml). Means ± SD, $n = 3$, are shown.

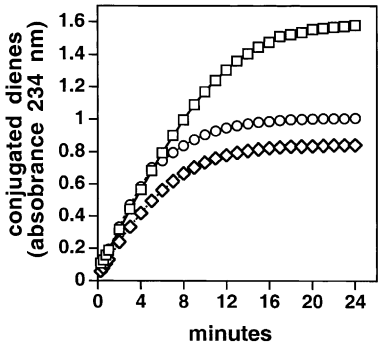


Table 1: Initial rates of lipoxygenase-mediated oxidation of phospholipid

	Conjugated dienes (nmoles/min/ml)	P values
no peptide (n=9)	4.44±1.13	
167-R173C-184 (n=7)	3.16±0.66	0.014
145-R151C-162 (n=6)	3.22±0.83	0.032

Values of conjugated dienes are Means±SD. The concentration of each peptide was 400 μg/ml. P values referr to "no peptide" comparisons and were calculated using Student's two-tailed test adjusting for sample sizes.

Fig. 4. Characteristic of synthetic peptides inhibiting lipoxygenase- mediated lipid peroxidation. The line graph on the left shows that peptide 145-R151C-162 inhibited the oxidation of phospholipid when added before (diamonds) and after (circles) initiation with soybean lipoxygenase (5 U/μl). Table 1 shows that peptides 167-R173C-184 and 145-R151C-162 inhibited the initial rate of lipid peroxidation. The values listed are a composite of all the data obtained using 400 μg/ml of each peptide.

peptide 167-R173C-184 (data not shown). Both synthetic peptides inhibited the initial rate of lipid peroxidation (Fig. 4, Table 1) similar to the behavior of apoA-I_{Milano} that we previously reported [14]. However, our previous studies demonstrated that apoA-I_{Paris} was unable to inhibit initial rates catalyzed by lipoxygenase indicating the two full-length cysteine variants differed in antioxidant potential. This difference is not evident using the synthetic peptides, indicating that contextual constraints limit antioxidant activity when the free thiol is positioned within helix 6 of apoA-I_{Paris}.

Cysteine-free peptides based on helix 7 (aa 167–184) and helix 6 (aa 145–162) of apoA-I_{WT} were weak inhibitors of lipid peroxidation, as shown in Fig. 5. In Figs. 5A and B, lipid peroxidation was initiated using soybean lipoxygenase. As can be seen, both peptides which lacked cysteine residues failed to inhibit initial rates of lipid peroxidation while only modest protection was achieved at latter times (>10 min) when lipid peroxides accumulated to relatively high levels (~50 μM) consistent with the behavior of apoA-I_{WT} [14]. Figs. 5C and D show that cysteine-free peptides were poor antioxidants in response to superoxide anions generated via xanthine/xanthine oxidase. The peptide based on helix 6 of apoA-I_{WT} appeared to be slightly better than peptides 167–184 in protecting phospholipid from ROS. This may be related to the presence of a single methionine residue (M148) in peptides 145–162. Methionine residues in apoA-I_{WT} have been shown to buffer against lipid peroxide accumulation [22]. However, our results using synthetic peptides as well as the full-length cysteine variants indicate that the presence of a free cysteine residue imparts more substantial antioxidant activity compared to methionine residues.

The cysteine residue in apoA-I_{Milano} and apoA-I_{Paris} is positioned at the polar/nonpolar interface of their respective amphipathic α helices. The position of the free thiol located near the lipid–water interface may account for the antioxidant activity directed toward lipid

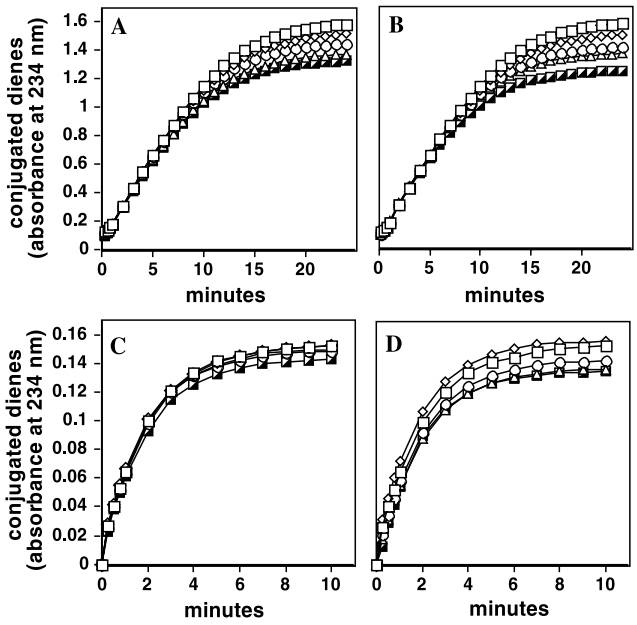


Fig. 5. Antioxidant activity of cysteine-free "control" peptides. The top panels (A, B) show protection against lipoxygenase (5 U/μl) mediated oxidation of phospholipid and bottom panels (C, D) protection from ROS generated via X/Xo (20 U/ml). Panels (A, C) Antioxidant activity of the helix 7 peptide (aa 167–184) derived from apoA-I_{WT}. Panels (B, D) Antioxidant activity of peptide (aa 145–162) derived from helix 6. Open squares represent the full extent of oxidation in the absence of peptide; diamonds, circles, triangles, and half-darkened squares correspond to 100, 200, 300, and 400 μg protein/ml, respectively. Results are representative from a total of four separate experiments; two experiments conducted with lipoxygenase and two using xanthine/xanthine oxidase.

surfaces, but the predicted water accessibility of thiol may also allow important interaction with water-soluble antioxidants, such as reduced glutathione. We have found that GSH acted synergistically, enhancing the antioxidant activity of apoA-I cysteine variants [14]. The results in Fig. 6A show that GSH can enhance the ability of peptide 167-R173C-184 to inhibit lipid

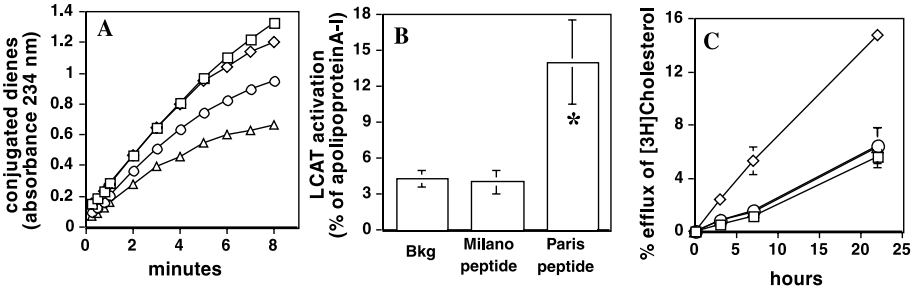


Fig. 6. Functional properties of peptide mimetics. (A) Interaction between peptide 167-R173C-184 with reduced glutathione (GSH, 100 μM). Squares show oxidation of phospholipid micelles exposed to soybean lipoxygenase (5 U/μl); diamonds, circles, and triangles correspond to GSH alone, the peptide alone (200 μg/ml), and GSH plus the peptide, respectively. (B) Activation of LCAT. Values are relative to the full extent of activation obtained in the presence of apoA-I_{WT}. Bkg = background values in the absence of enzyme. Means ± SD, $n = 3$. (C) Cholesterol efflux from J774 macrophages to lipid-free apoA-I_{WT} (diamonds), the apoA-I_{Milano} peptide (cirles), and the peptide mimetic of apoA-I_{Paris} (squares). Triangles behind squares and circles correspond to nonspecific efflux to serum-free medium. Protein concentrations were 100 μg/ml. Means ± SD, $n = 3$, are shown.

peroxidation, consistent with previous results using the full-length cysteine variants. Similar results were obtained using the peptide 145-R151C-162 (data not shown). Helix 6 of apoA-I_{WT} possesses a cluster of three positively charged arginine residues (R149, R153, and R160) located just off its lipid–water interface that are highly conserved and important in LCAT activation [23], whereas helix 7 lacks this series of arginine residues. The thiol-bearing synthetic peptide 145-R151C-162 based on helix 6 was found to activate LCAT as shown in Fig. 6B, but the peptide based on helix 7 could not activate the enzyme consistent with the role of arginine residues in LCAT activation. Helix 6 and 7 peptides based on apoA-I_{Paris} and apoA-I_{Milano} failed to promote cholesterol efflux from J774 macrophages (Fig. 6C), consistent with previously published studies using isolated helical segments of apoA-I_{WT} [24].

Discussion

In this study, we found that the monomeric forms apoA-I_{Milano} and apoA-I_{Paris} inhibited the initiation/amplification of phospholipid oxidation induced by water-soluble ROS, i.e., superoxide anion and hydroxyl radicals. This activity was not associated with apoA-I_{WT}, indicating that the presence of the free thiol in the apoA-I cysteine variants conferred a gain-of-function. Yet, the lipid-free forms of apoA-I_{Milano} and apoA-I_{Paris} were unable to directly quench ROS in aqueous solution. This indicates that the apoA-I cysteine variants acted at the level of phospholipid to inhibit oxidation. Moreover, the results using synthetic peptides rule out the possibility that a conformational change associated with the lipidation of the apolipoprotein variants was needed to directly quench water-soluble free radicals. The synthetic peptides bearing a single thiol residue effectively inhibited lipid peroxidation, but they also failed to quench superoxide anions. The peptides are capped at the ends and do not require unfolding to form amphipathic α helices, indicating that the inherent nature of the peptides, like the full-length cysteine variants, is to specifically quench lipid peroxides on phospholipid surfaces.

These studies using the monomeric forms of cysteine variants as well as their synthetic peptide mimetics provide valuable insights into the mechanism by which apoA-I_{Milano} may protect against CVD, as suggested by clinical studies [5,6,8]. The observation that apoA-I_{Milano} possessed antioxidant activity directed toward phospholipid surfaces indicates that the monomeric form of the cysteine variant may act in vivo on the surface of lipoproteins particles, such as HDL, to inhibit lipid peroxidation. The presence of monomeric and dimeric forms of apoA-I_{Milano} on the surface of HDL may act together to ensure efficient removal of cell-derived cholesterol under conditions of oxidative stress. Calabrasi et al. [12,25] suggested that dimerization of apoA-I_{Milano} may be associated with the formation of a structural domain that protrudes from the surface of small HDL particles thereby enhancing plasma membrane interactions and facilitating cholesterol efflux via aqueous diffusion. It is known that the phospholipid on the surface of HDL is important for efficient efflux of cell-derived cholesterol via the aqueous diffusion mechanism and that oxidation of phospholipid inhibits cholesterol efflux to HDL [26]. Thus, the monomeric form apoA-I_{Milano} protecting phospholipid from oxidation may allow subfractions of small, dense HDL particles to effectively participate in the initial stages of reverse cholesterol transport that require an intact phospholipid surface for solubilizing cell-derived cholesterol. In support of this suggestion, the smallest of HDL_3 particles present in apoA-I_{Milano} carriers appears to be enriched in the monomeric form of the variant [10].

It is now thought that lipid-free apoA-I_{Milano} and apoA-I_{Paris} are just as effective as apoA-I_{WT} in promoting cholesterol efflux from cells via the specific apo-

lipoprotein-mediated pathway involving the ABCA1 transporter [13,14]. Since the antioxidant activity of the apoA-I cysteine variants is directed toward phospholipid surfaces, it is reasonable to speculate that such activity is manifested upon the lipidation of the variants in vivo. The ABCA1 transporter is regulated by a PPAR-γ/LXR-α transcription pathway in response to lipid mediators ingested upon the uptake of oxidized lipoproteins by macrophages [27,28]. Cholesterol enrichment of macrophages also upregulates ABCA1 gene expression, leading to enhanced lipid efflux to lipid-free apolipoproteins [29]. This suggests that the antioxidant activity of apoA-I_{Milano} is likely to be concentrated where it is needed most in metabolically active sites of inflammation and cholesterol deposition in the artery wall where there is an upregulation of the ABCA1 transporter. As such, it may be possible to target a therapeutic, in the form of a synthetic peptide, to sites of inflammation to provide a high degree of protection from atherogenic events, thus, preventing further rounds of cholesterol deposition.

In this study, we report for the first time that thiol-bearing synthetic peptides (18 mer) retained the antioxidant activity of apoA-I_{Milano}; whereas, cysteine-free peptides based on helices 7 and 6 of apoA-I_{WT} were relatively weak inhibitors of lipid peroxidation. These results support our hypothesis that incorporation of a free thiol within amphipathic α helices is necessary to confer the antioxidant activity of apoA-I_{Milano}. We demonstrate that a synthetic peptide based on amino acids 167-R173C-184 exhibited the antioxidant characteristics of the full-length cysteine variants. These characteristics include thiol-dependent antioxidant activity directed toward phospholipid surfaces as well as synergistic interactions with reduced glutathione. The latter may be important in vivo, increasing the antiatherogenic capacity of the variants. The only difference that we observed was that higher doses of the 18 mers were required to inhibit lipid peroxidation compared to apoA-I_{Milano}. However, this is to be expected as the full-length cysteine variant possess, a high affinity for lipid surfaces while it is known that helices 7 and 6 of apoA-I, in isolated form as synthetic peptides, exhibit relatively low affinity for phospholipid surfaces [24,30].

Synthetic peptides based on apoA-I_{Milano} and apoA-I_{Paris} exhibited the same capacity to inhibit oxidative events in contrast to the full-length cysteine variants, which differed in antioxidant activity. Amino acids 167–184 corresponding to the segment harboring the R173C mutation, and the stretch of amino acids, i.e. 145–162, containing R151C both form class A amphipathic α helices in which positively charged arginine residues are located at the lipid–water interface and negatively charged residues are positioned in the middle of the polar face of the helix. The nonpolar face of each helix is relatively small consisting of four hydrophobic residues such as leucine, alanine and valine residues. The R $\rightarrow$ C substitution is located at the same position at the polar/nonpolar interface of each helical segment. It is therefore not surprising that the synthetic peptides based on apoA-I_{Milano} and apoA-I_{Paris} exhibited the same capacity to inhibit lipid peroxidation as the peptides appear quite similar with regard to their overall structure, although their primary amino acid sequences are quite different. This suggests that the difference in antioxidant activity between the two, full-length cysteine variants is related to contextual factors within the apolipoprotein molecule that optimize antioxidative behavior when the free thiol is positioned within helix 7 as is the case with apoA-I_{Milano}. Helix 7 possesses a rather low lipid-binding affinity in isolated form compared to the other helical repeats including adjacent helical segments, which bind lipid with slightly greater affinity [24,30]. This arrangement of alternating lipid affinities may dictate the behavior of helical segments present within the context of the entire C-terminal lipid-binding domain of apoA-I, thus, maximizing the frequency of collision between the free thiol at position R173C with reactive lipid peroxides on phospholipid surfaces.

The apoA-I_{Milano} and apoA-I_{Paris} mutations are located in helical segments that have been implicated in LCAT activation [31–34]. Dimerization of apoA-I_{Milano} has been shown to impair this activation step reducing cholesterol esterification [35]. Yet, indirect evidence obtained in transgenic mice expressing apoA-I_{Milano} in the absence of apoA-I_{WT} suggests that the monomeric form of the variant can activate LCAT normally, as determined by endogenous cholesterol esterification rates [36]. The free thiol in helix 6 of apoA-I_{Paris} is located just opposite a series of three interfacial arginine residues that are highly conserved and important for LCAT interactions, leading to enzyme activation [23]. Helix 7 lacks this cluster of arginine residues. The synthetic peptide 145-R151C-162 activated the enzyme while the peptide based on helix 7 failed in this regard. However, the helix 6 peptide was only partially active indicating that the cluster of three arginine residues was not sufficient, in and of itself, to mediate complete activation of the enzyme. This suggests that cooperative interactions between adjacent amphipathic α helices are required to fully activate LCAT, as suggested in previous studies [31–34].

The present studies provide evidence that the presence of a cysteine residue within a synthetic amphipathic α helical peptide confers a unique biological activity that is associated with the monomeric form of apoA-I_{Milano}. This represents an important first step in the development of therapeutic agents based on the newly discovered activity of the apoA-I_{Milano}. It is generally thought that free-radicals in the artery wall promote the oxidation of phospholipid generating fragmented products that are pro-inflammatory mediators of atherogenesis

[37]. The synthetic peptides used in the present studies were found to possess an antioxidant activity on phospholipid surfaces just like the full-length cysteine variant. As the inherent nature of apoA-I_{Milano} and its peptide mimetics is to act on the surface of phospholipids to inhibit lipid peroxidation, it is reasonable to speculate that synthetic peptides that exhibit a high affinity for lipid surfaces may inhibit the oxidation of phospholipid at concentrations more like the full-length variant. An important question is whether addition of a cysteine residue to different helical repeats of apoA-I, instead of helices 7 and 6, confers the antioxidant activity of apoA-I_{Milano}. The use of different amphipathic α helices may allow the development of therapeutic peptides that share native apoA-I properties in mediating ABCA1 interactions and promoting cellular cholesterol efflux in addition to the antioxidant characteristic of apoA-I_{Milano}.

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